

# Bode Plotter PCB

Swept Frequency Analyser · 10 mHz to 100 MHz

**10 mHz**  
to 100 MHz

**$\pm 30$  dB**  
Magnitude

**$\pm 1^\circ$**   
Phase accuracy

**~₹8k**  
All-in cost

EE24BTECH11003: Akshara Sarma · EE24BTECH11049: Faraz P · EE24BTECH11059: Siddhanth

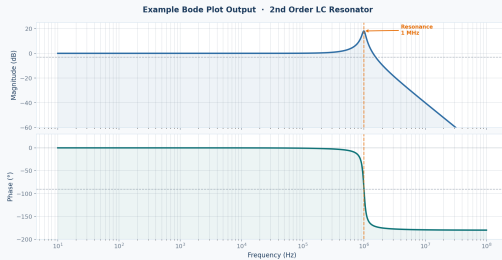
AD9913 DDS · AD8302 · STM32H750VBT6 · March 2026

# The Problem

Engineers need frequency response data.

Today's options:

- ▶ Lab VNA — accurate but ₹20k - ₹2 Lakhs
- ▶ Bench scope + sig gen — slow, manual, error-prone
- ▶ NanoVNA — cheap but PC-tethered, limited range



**None are self-contained, affordable & embedded.**

# A standalone, embedded Bode plotter for ₹8k.

Sweep 10 mHz to 100 MHz. Read magnitude and phase on a 2.4 inch display. Log results to SD card. No PC. No software. Just power and a DUT.

**Self-contained**

No PC needed

**Open firmware**

Full STM32 control

**Logs to SD**

CSV export

**~₹8k built**

4-layer PCB

# Who Is It For

## Embedded Engineers

Test filters and amplifiers during hardware bring-up without leaving the bench.

## University Labs

Affordable teaching tool for circuits and signals courses.  
Student-buildable.

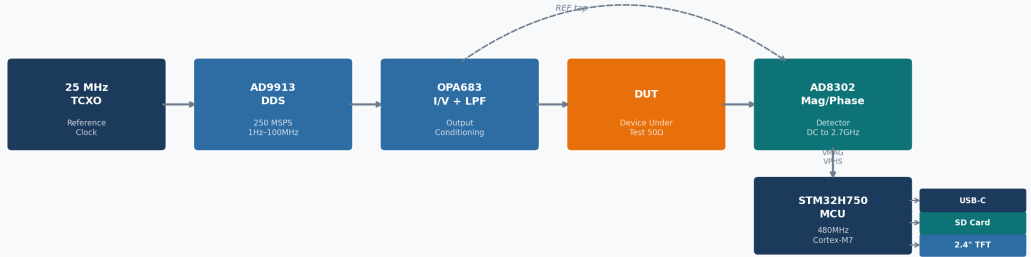
## RF Hobbyists

Characterise hand-wound inductors, ceramic filters, antenna matching networks.

Anyone who needs frequency response data and doesn't want to spend ₹50000 on a VNA.

# How It Works

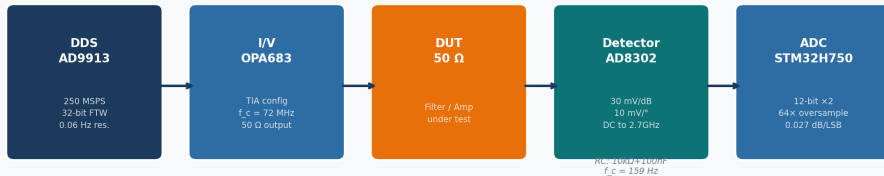
System Block Diagram



DDS generates swept sine · AD8302 measures ratio of output to input · STM32 reads two DC voltages and plots the result in real time

# Signal Chain

## Signal Chain — Stage by Stage



### Stimulus

AD9913 DDS sweeps from  $f_{\text{start}}$  to  $f_{\text{stop}}$  in log-spaced steps. 32-bit tuning word: 0.058 Hz

### Measurement

AD8302 outputs two calibrated DC voltages: VMAG (30 mV/dB) and VPHS (10 mV/°).

### Processing

STM32 ADC reads both channels with 64x oversampling, converts to dB/°, updates the display.

# Three chips do all the work.

## AD9913

DDS Synthesiser

- ▶ 250 MSPS system clock
- ▶ 10 mHz to 100 MHz output
- ▶ 32-bit frequency word
- ▶ Built-in linear sweep mode
- ▶ SPI interface

## AD8302

Gain/Phase Detector

- ▶ DC to 2.7 GHz input range
- ▶  $\pm 30$  dB magnitude range
- ▶  $0^\circ$  to  $180^\circ$  phase range
- ▶  $30 \text{ mV/dB} \cdot 10 \text{ mV/}^\circ$  output
- ▶ Single supply, 5 V

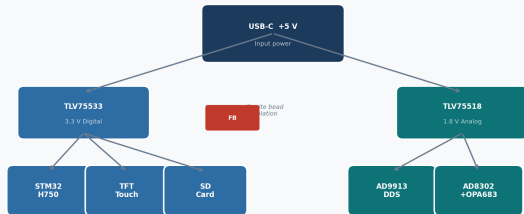
## STM32H750

MCU & Controller

- ▶ 480 MHz Cortex-M7
- ▶ 12-bit ADC,  $64\times$  oversample
- ▶ SPI  $\times 4$ : DDS, TFT, SD, touch
- ▶ USB-C full speed
- ▶ LQFP-100, hand-solderable

# Power Architecture

Power Architecture



Both LDOs rated 1 A. Total system current: ~300 mA at full speed. Powered from any USB-C source.

## Two rails from USB-C +5 V:

- ▶ **TLV75533** → 3.3 V digital  
STM32, TFT, SD card, USB
- ▶ **TLV75518** → 1.8 V analog  
AD9913, AD8302
- ▶ **Ferrite bead** isolates VDDA  
Clean ADC reference, no digital noise on analog rail
- ▶ **1.8 V split:** analog vs. digital  
Separate pour planes, star-point GND



# Unique Selling Points

## No PC required

Full sweep, display and SD logging run on the STM32. Results appear on the 2.4in TFT in real time. Touch screen lets you read off frequency, gain and phase at any point.

## Open firmware

STM32H750 with full HAL access. Change sweep range, point count, settle time, add USB streaming, add Wi-Fi module — all in C.

## 10 mHz floor

AD9913 32-bit FTW gives 0.06 Hz resolution. Most comparable tools start at 50 kHz or higher.

# Cost Breakdown

| Item                        | Cost           |
|-----------------------------|----------------|
| STM32H750VBT6               | ₹665.71        |
| AD9913BCPZ (DDS)            | ₹2,722.15      |
| AD8302ARUZ                  | ₹3,112.06      |
| OPA683ID                    | ₹468.96        |
| 2.4" TFT + ILI9341          | ₹500           |
| LDOs, passives, connectors  | ≈  500         |
| crystal, etc.               | ₹300           |
| <b>All-in (single unit)</b> | <b>~₹8,270</b> |

Comparable lab VNA: ₹20k - 13L

This instrument: ₹8k

## Specifications at a glance

| Parameter        | Value                     | Feature  | Detail             |
|------------------|---------------------------|----------|--------------------|
| Frequency range  | 10 mHz – 100 MHz          | Display  | 2.4" TFT, touch    |
| Magnitude range  | $\pm 30$ dB (60 dB total) | Storage  | microSD, CSV       |
| Phase range      | $0^\circ - 180^\circ$     | USB      | USB-C Full Speed   |
| Mag. accuracy    | $\pm 0.5$ dB              | Power    | USB-C +5 V         |
| Phase accuracy   | $\pm 1^\circ$             | MCU      | STM32H750, 480 MHz |
| Freq. resolution | 0.058 Hz                  | DDS      | AD9913, 250 MSPS   |
| Sweep points     | Up to 4096                | Detector | AD8302, DC–2.7 GHz |
| Input impedance  | $50\ \Omega$              | Build    | Hand-solderable    |

# Roadmap

## Rev 1.0 — Now

AD9913 + AD8302 + STM32H750  
2.4" TFT · SD card · USB-C  
10 mHz–100 MHz ·  $\pm 30$  dB ·  $\pm 1^\circ$   
₹8k all-in

## Rev 2.0 — Near Term

Provide complete DUT analysis  
Increase freq range upto 1GHz  
Increase gain range upto  $\pm 50$ dB-60dB

Each revision remains hand-solderable and under ₹10000 built.

**NEXT STEPS**